LEARNER GUIDE

Estimating and judging reasonableness

Number Systems and Calculator Judgement · Lesson 7. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

To estimate, round each number to **1 significant figure** (its first non-zero digit) and work out the easier calculation. Under a √, round to a nearby square number. A good estimate tells you whether an answer is about right. It can also tell you which way it leans: rounding a number **up** pushes a product up, but rounding a number you **divide by** up pulls the answer down. If every rounding pushes the same way, you know you have an **overestimate** or an **underestimate**.

What you are learning to notice

- The first significant figure is the first non-zero digit: 0.0372 ≈ 0.04, and 9.81 ≈ 10.
- √50 lies between √49 = 7 and √64 = 8, much nearer 7. Square a guess to test it: $7.1^2 = 50.41$.
- Dividing by a number less than 1 makes the answer bigger: ÷ 0.5 is the same as × 2.
- Numerator rounded up and denominator rounded down: definitely an overestimate. If the roundings push different ways, you cannot be sure.
- An answer about 10 times too big or too small often means a decimal point slipped, or the calculation was keyed wrongly.

Moving through it, one step at a time

Round each number to 1 significant figure (or to a nearby square number under a √). Work out the estimate. Compare it with the answer: about the same size, or way out? Then mark each rounding with an arrow, ↑ or ↓, for the way it pushes the answer, and decide: over, under, or cannot be sure.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — The sense detector

Someone says 812 ÷ 3.9 = 20.82. Is that sensible?

1. Round each number to 1 significant figure.	812 → 800, 3.9 → 4
2. Estimate.	800 ÷ 4 = 200
3. Compare: 20.82 is about 10 times smaller than 200.	not sensible
4. Direction: 812 rounded down ↓; 3.9 rounded up, and we divide by it ↓.	an underestimate

20.82 is not sensible: the answer is just over 200 (it is 208.2…).

Example 2 — Locating a root

Estimate √50 to 1 decimal place.

1. Find the squares either side.	$7^2 = 49$, $8^2 = 64$
2. 50 is much nearer 49.	√50 is just over 7
3. Test a guess by squaring it.	$7.1^2 = 50.41$
4. 50 is between $7.0^2 = 49$ and $7.1^2 = 50.41$, and much nearer 50.41.	√50 ≈ 7.1

√50 ≈ 7.1 to 1 decimal place.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (rounding, square numbers, and estimating before you calculate), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.

- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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